Poll Everywhere Quizzes

At the beginning of each lecture:

1. Use your device to log in to PollEverywhere (PE). Enter your ANU email (uXXXXXXX@anu.edu.au); PE supports ANU SSO.

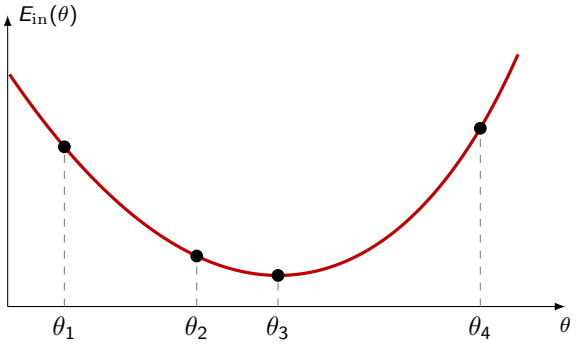
   https://pollev.com/anuiml2026



Photographing or recording the quiz questions is not permitted

Lecture 11 - Quiz 1 (120 seconds)

L11Q1 - The plot shows the in-sample error $E_{\text{in}}(\theta)$ of a model with one parameter. One step of the update $\theta \leftarrow \theta - \alpha\, E'_{\text{in}}(\theta)$ is taken from each of the four marked values, with the **same** small $\alpha > 0$. From which value does the parameter move **furthest to the right**?

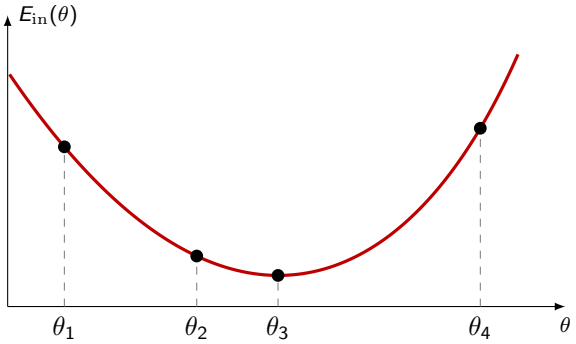


- A) From θ_1
- B) From θ_2
- C) From θ_3
- D) From θ_4

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L11Q1 - The plot shows the in-sample error $E_{\text{in}}(\theta)$ of a model with one parameter. One step of the update $\theta \leftarrow \theta - \alpha\, E'_{\text{in}}(\theta)$ is taken from each of the four marked values, with the **same** small $\alpha > 0$. From which value does the parameter move **furthest to the right**?

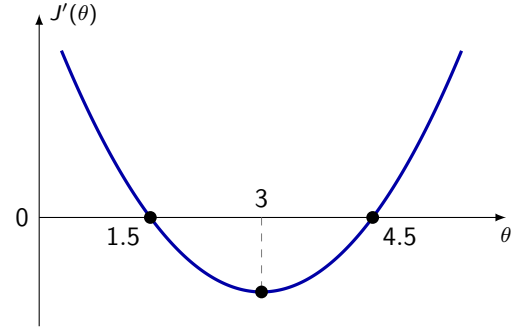


- **A) From** θ_1 *The step is* $-\alpha\, E'_{\text{in}}(\theta)$*: to the right where the derivative is negative, and longer where the curve is steeper. At* θ_3 *the derivative is zero and nothing moves. From* θ_4 *the step is the longest of all, but to the left.*
- B) From θ_2
- C) From θ_3
- D) From θ_4

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Lecture 11 - Quiz 2 (120 seconds)

L11Q2 - The plot shows the **derivative** $J'(\theta)$ of an error $J(\theta)$, and not the error itself. At which value of θ does the error J have a **local minimum**?

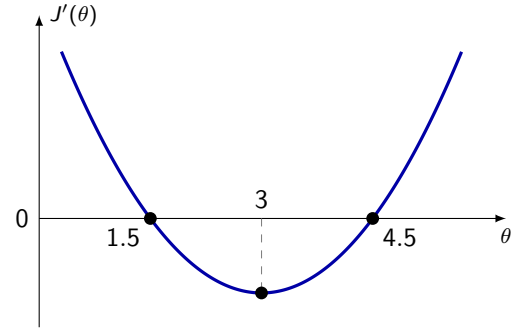


- A) At $\theta = 3$, the lowest point of the plotted curve
- B) At $\theta = 1.5$, where the derivative is zero and decreasing
- C) At $\theta = 4.5$, where the derivative is zero and increasing
- D) At both $\theta = 1.5$ and $\theta = 4.5$, since the derivative is zero at both

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Lecture 11 - Quiz 2 (120 seconds)

L11Q2 - The plot shows the **derivative** $J'(\theta)$ of an error $J(\theta)$, and not the error itself. At which value of θ does the error J have a **local minimum**?



- A) At $\theta = 3$, the lowest point of the plotted curve
- B) At $\theta = 1.5$, where the derivative is zero and decreasing
- **C) At $\theta = 4.5$, where the derivative is zero and increasing** *There J' goes from negative to positive: J decreases, then increases. At 1.5 it is the opposite, a local maximum. At 3, J decreases fastest.*
- D) At both $\theta = 1.5$ and $\theta = 4.5$, since the derivative is zero at both

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